

100 Days of Linux - Day 026

Title: Real-World Data Engineering Search Practice

Today's Goal

Strengthen your file searching skills using realistic data engineering folders and datasets.

Why This Matters

Data engineers constantly search for CSV files, log files, SQL scripts, and configuration files inside large project directories.

Commands Reviewed Today

find, grep, wildcards (*, ?, []), ls, pwd, mkdir, touch, nano

Beginner Lesson

Today you'll work with a small project structure similar to a real data engineering project. Your focus is to locate files quickly and search for useful information inside them.

Practice Questions

1. Create a folder named retail_project with subfolders data, logs, and scripts. Inside data create employees.csv, customers.csv, and sales.csv.
2. Using Nano, add at least 5 rows of sample data to employees.csv. Use find to locate employees.csv.
3. Create application.log inside the logs folder. Add the word ERROR on one line and INFO on another. Use grep to search for ERROR.
4. Use a wildcard to list only all .csv files inside the data folder.
5. Display your current working directory and list every directory inside retail_project.

Hints

Remember: find searches for files, grep searches inside files, and *.csv matches every CSV file.

Mini Challenge

Create orders.csv, products.csv, and pipeline.log. Use find to locate every CSV file, grep to search for a customer name, and verify your folder structure with ls.

Common Mistakes

Using grep to search for filenames. Use find for filenames and grep for text inside files.

Did You Know?

Many ETL pipelines process hundreds of CSV files every day,

making search commands essential.

Pro Tip

Organize projects into folders like data, logs, scripts, and config to make navigation easier.

Answer Key (Instructor Only)

Q1: mkdir retail_project && cd retail_project && mkdir data logs scripts && touch data/employees.csv data/customers.csv data/sales.csv

Q2: nano data/employees.csv && find . -name 'employees.csv'

Q3: nano logs/application.log && grep 'ERROR' logs/application.log

Q4: ls data/*.csv

Q5: pwd && find retail_project -type d

Homework

Create another project named insurance_project with CSV and log files, then practice find, grep, and wildcards.

Next Day Preview

Tomorrow is Revision Quiz #3 covering find, locate, grep, and wildcards.